COVER PAGE

LAB 3
Submission

Name: Bisma Mazhar

Roll No: 44

Class: BSCS

Section: C

Submitted To:Sir Asad Majeed

Date: 16-10-2026

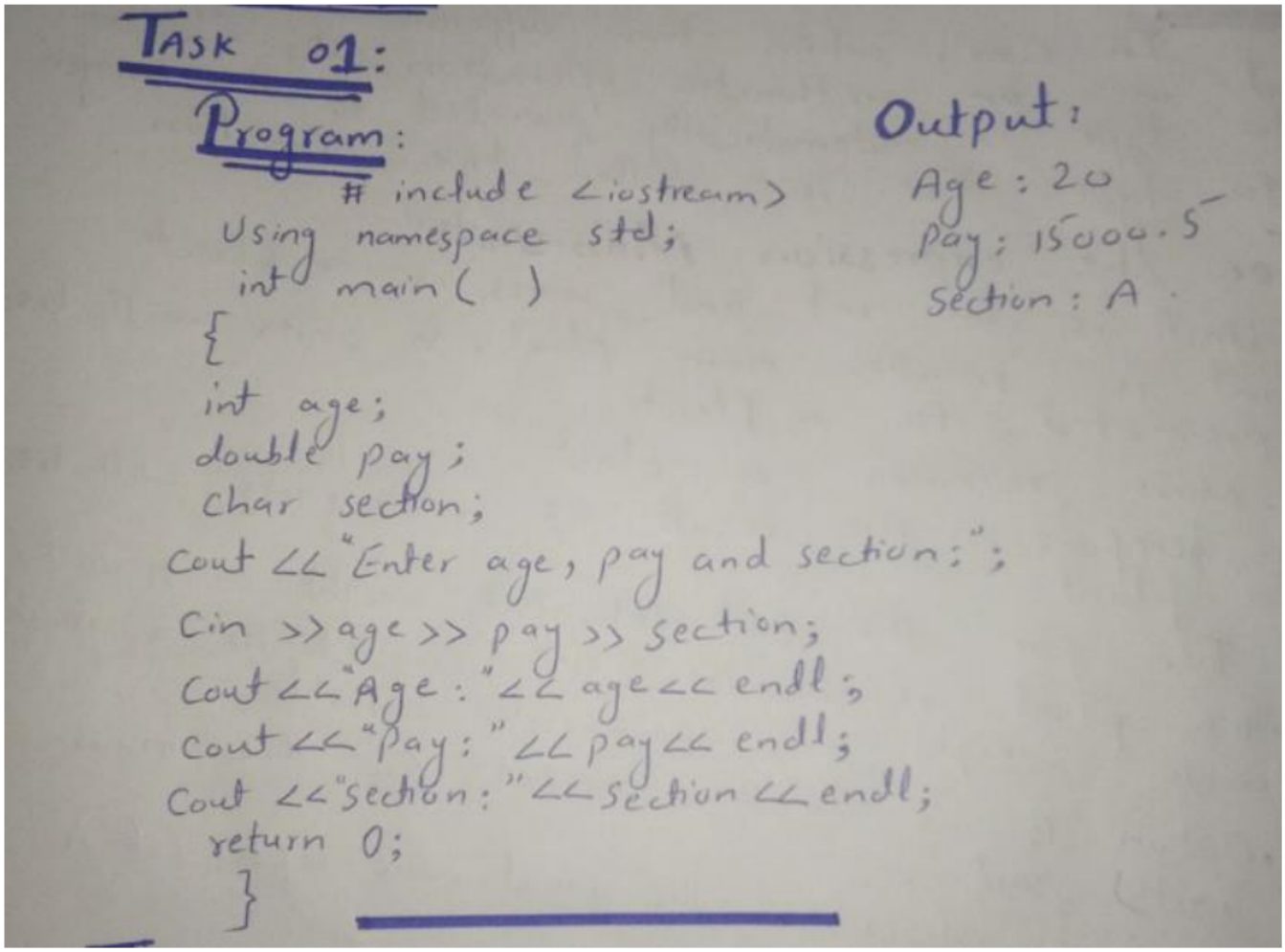
LAB TASK 1 — QUESTION & HANDWRITTEN CODE

LAb Task 1:

Question:
Task 1:
Task 1:
Assume that the following variables are defined:
int age;
double pay;
char section;

Write a single cin statement that will read input into each of these variables.

Handwritten code:



Code on compiler:

LAB TASK 1 — CODE & OUTPUT

Coding C++

Auto saved at 20:55:40 RUN MENU

```cpp
#include <iostream>
using namespace std;
int main()
{
    int age;
    double pay;
    char section;
    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section;
    cout << "Age: " << age << endl;
    cout << "Pay: " << pay << endl;
    cout << "Section: " << section << endl;
    return 0;
}
```

Output:

Compile Result

```
Enter age, pay and section: 20  15000.50 A
Age: 20
Pay: 15000.5
Section: A

[Process completed - press Enter]
```

LAB TASK 2 — QUESTION

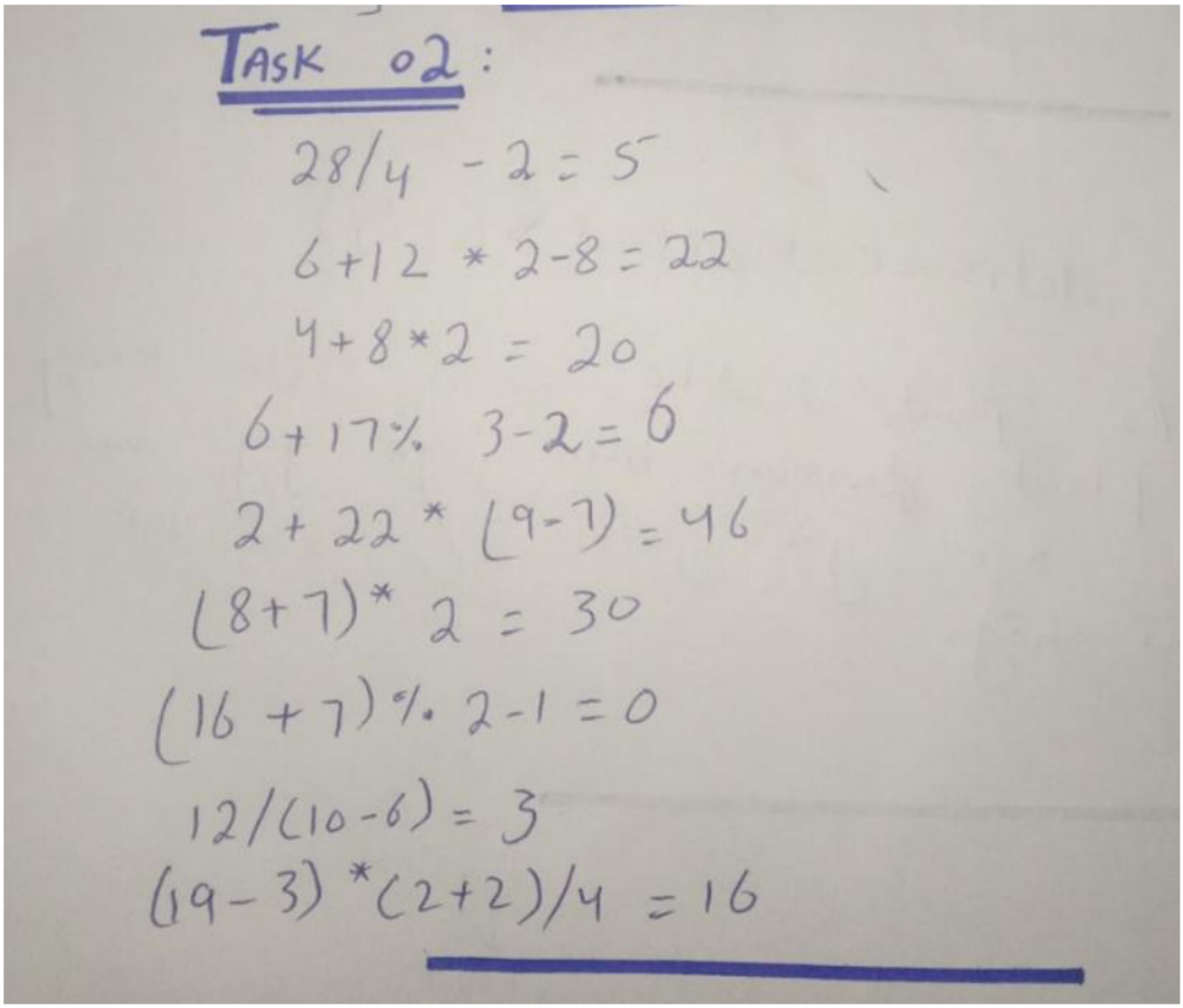
Lab Task 2

Question:
Task 2:
Complete the following table by writing the value of each expression in the Value column according to C++ language rules.

1. 28 / 4 - 2
2. 6 + 12 * 2 - 8
3. 4 + 8 * 2
4. 6 + 17 % 3 - 2
5. 2 + 22 * (9 - 7)
6. (8 + 7) * 2
7. (16 + 7) % 2 - 1
8. 12 / (10 - 6)
9. (19 - 3) * (2 + 2) / 4

Handwritten code:

LAB TASK 2 — HANDWRITTEN WORK



Code on compiler:

LAB TASK 2 — CODE / ANSWER

code

```
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
```

LAB TASK 3 — QUESTION

Lab Task 3:

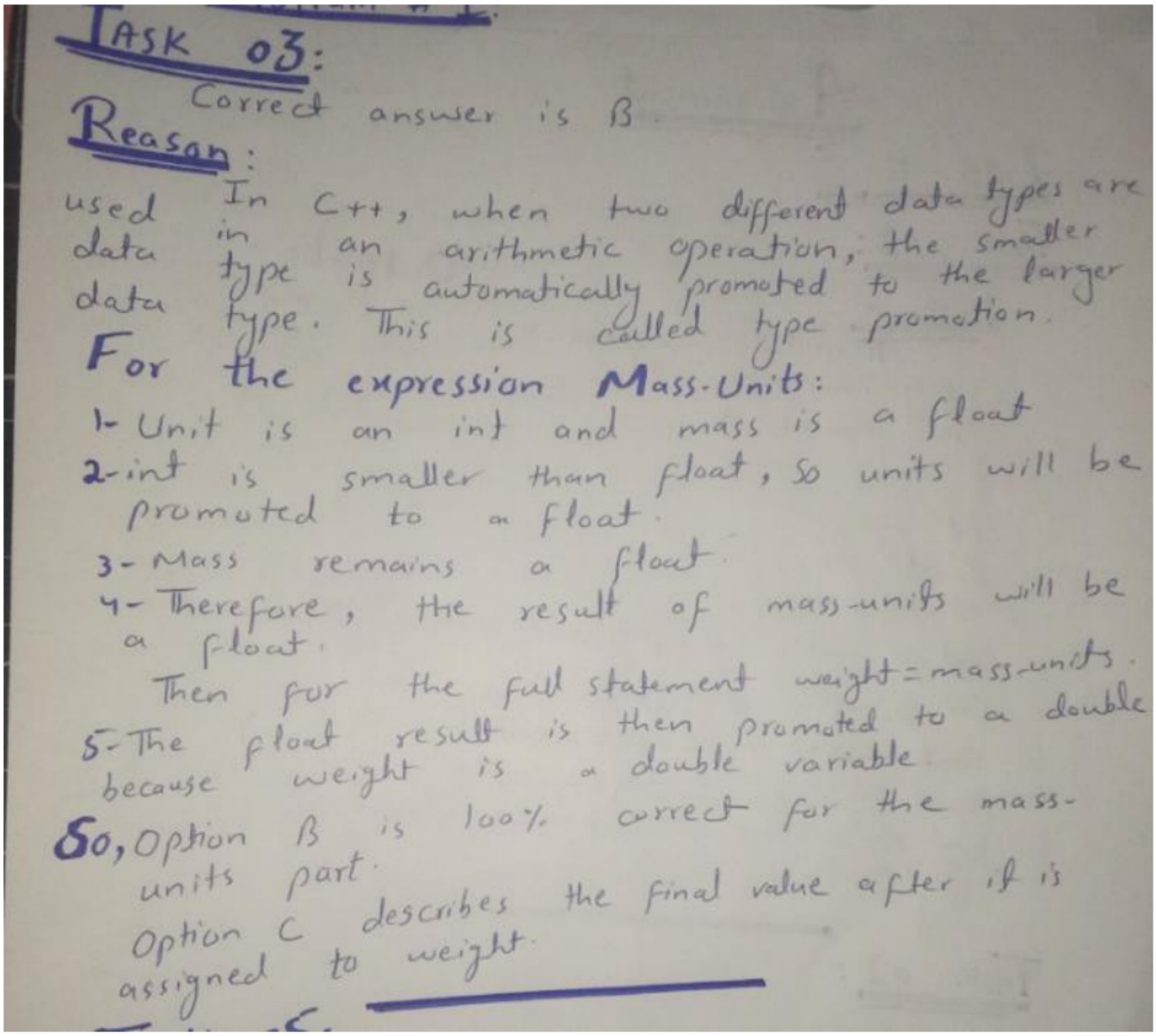
Question:
Task 3:
Task 3:
Assume a program has the following
variable definitions:
int units;
float mass;
double weight;
weight = mass * units;

Which automatic data type conversion will take place?

A. mass is demoted to an int, units remains an int, and the result of mass * units is an int.
B. units is promoted to a float, mass remains a float, and the result of mass * units is a float.
C. units is promoted to a float, mass remains a float, and the result of mass * units is a double.

Handwritten:

LAB TASK 3 — HANDWRITTEN WORK



Code on compiler:

LAB TASK 3 — ANSWER & EXPLANATION

Correct Answer is B

Reason:

In C++, when two different data types are used in an arithmetic operation, the smaller data type is automatically promoted to the larger data type. This is called **Type Promotion**.

For the expression `mass _ units`:

1. `units` is an `int` and `mass` is a `float`.
2. `int` is smaller than `float`, so `units` will be **promoted to a float**.
3. `mass` remains a `float`.
4. Therefore, the result of `mass _ units` will be a **float**.

Then for the full statement `weight = mass`

LAB TASK 4 — QUESTION & HANDWRITTEN CODE

Lab task 4

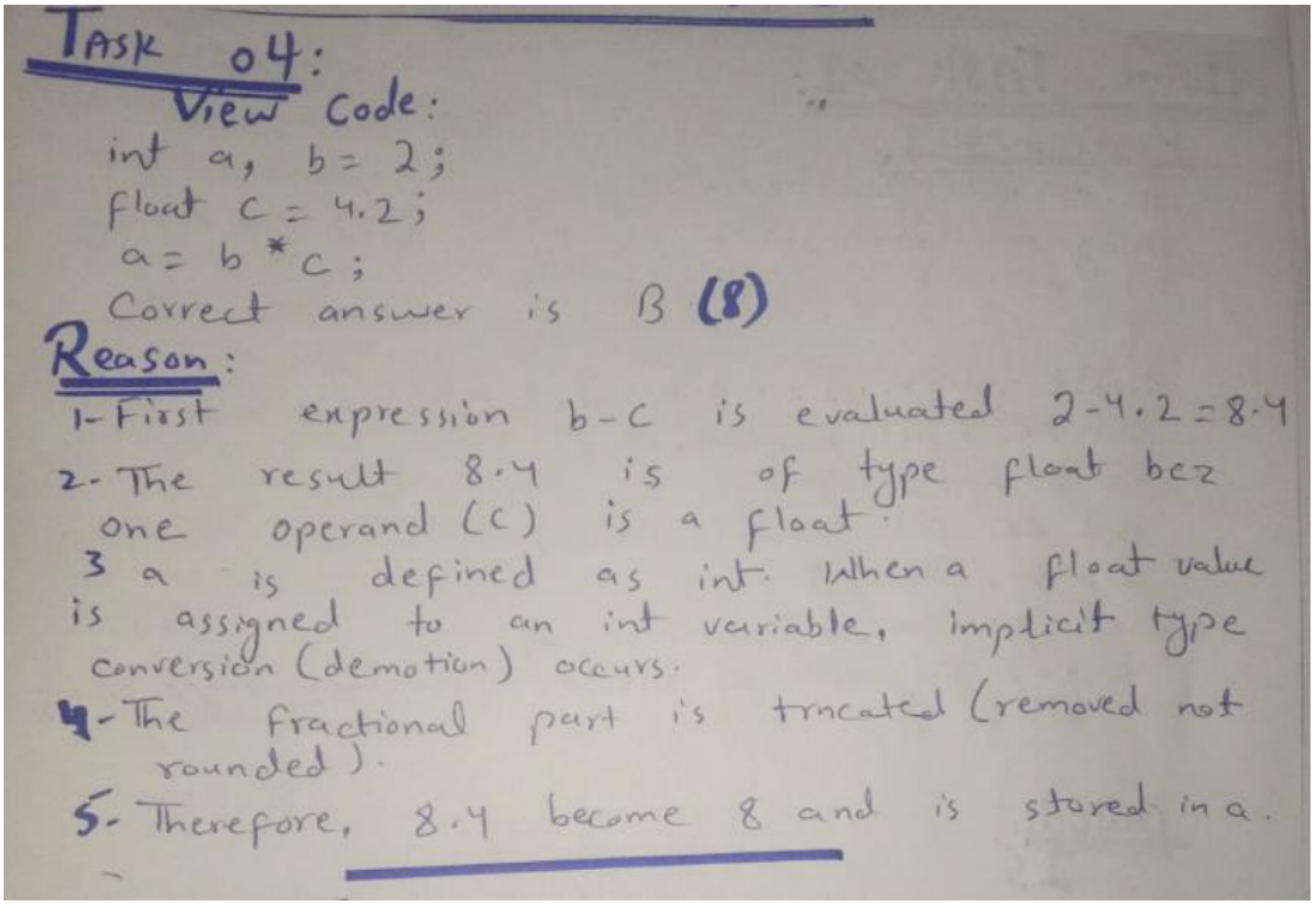
Question:
Task 4:
Assume a program has the following variable definitions:
int a, b = 2;
float c = 4.2;

And the following statement:
a = b * c;

What value will be stored in a?

A. 8.4
B. 8
C. 0
D. None of the above

Handwritten code:



Code on compiler:

LAB TASK 4 — ANSWER & EXPLANATION

```
int a, b = 2;
float c = 4.2;
a = b * c;
```

Correct Answer: B. 8

Reason in English:

1. First, the expression b * c is evaluated: 2 * 4.2 = 8.4
2. The result 8.4 is of type `float`, because one operand (c) is a float.
3. a is defined as an `int`. When a float value is assigned to an int variable, implicit type conversion (demotion) occurs.
4. The fractional part is truncated (removed), not rounded.
5. Therefore, 8.4 becomes 8 and is stored in a.

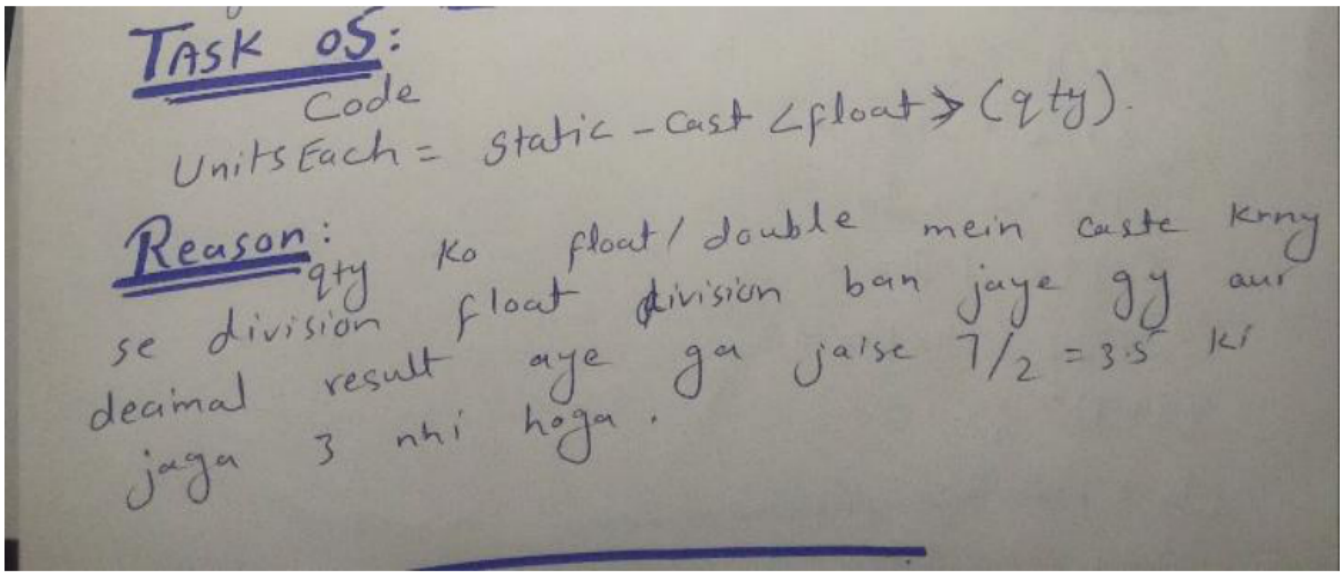
LAB TASK 5 — QUESTION & SOLUTION

Lab task 5

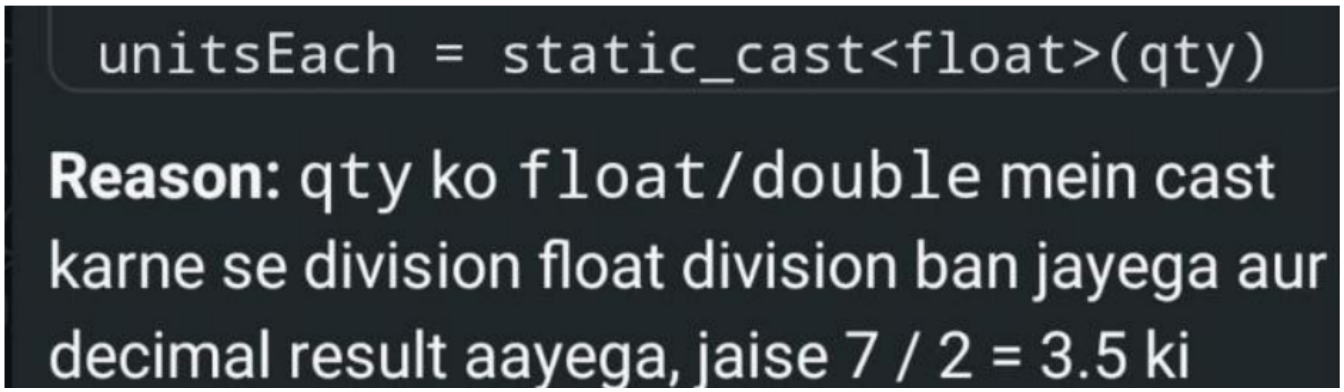
Question:
Task 5:
Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the following statement so it will no longer perform integer division.

unitsEach = qty / salesReps;

Handwritten:



Code on phone:



HOME TASK 1 — FIND THE ERRORS

Task 1:
Each of the following programs has some errors. Locate as many as you can.

Program-1

```
using namespace std;

void main ()
{
   double number1, number2, sum;

   cout << "Enter a number: ";
   Cin << number1;

   cout << "Enter another number: ";
   cin << number2;

   number1 + number2 = sum;

   cout << "The sum of the two numbers is " << sum
}
```

Program-2

```
#include <iostream>
using namespace std;

void main()
{
   int number1, number2;
   float quotient;

   cout << "Enter two numbers and I will divide\n";
   cout << "the first by the second for you.\n";

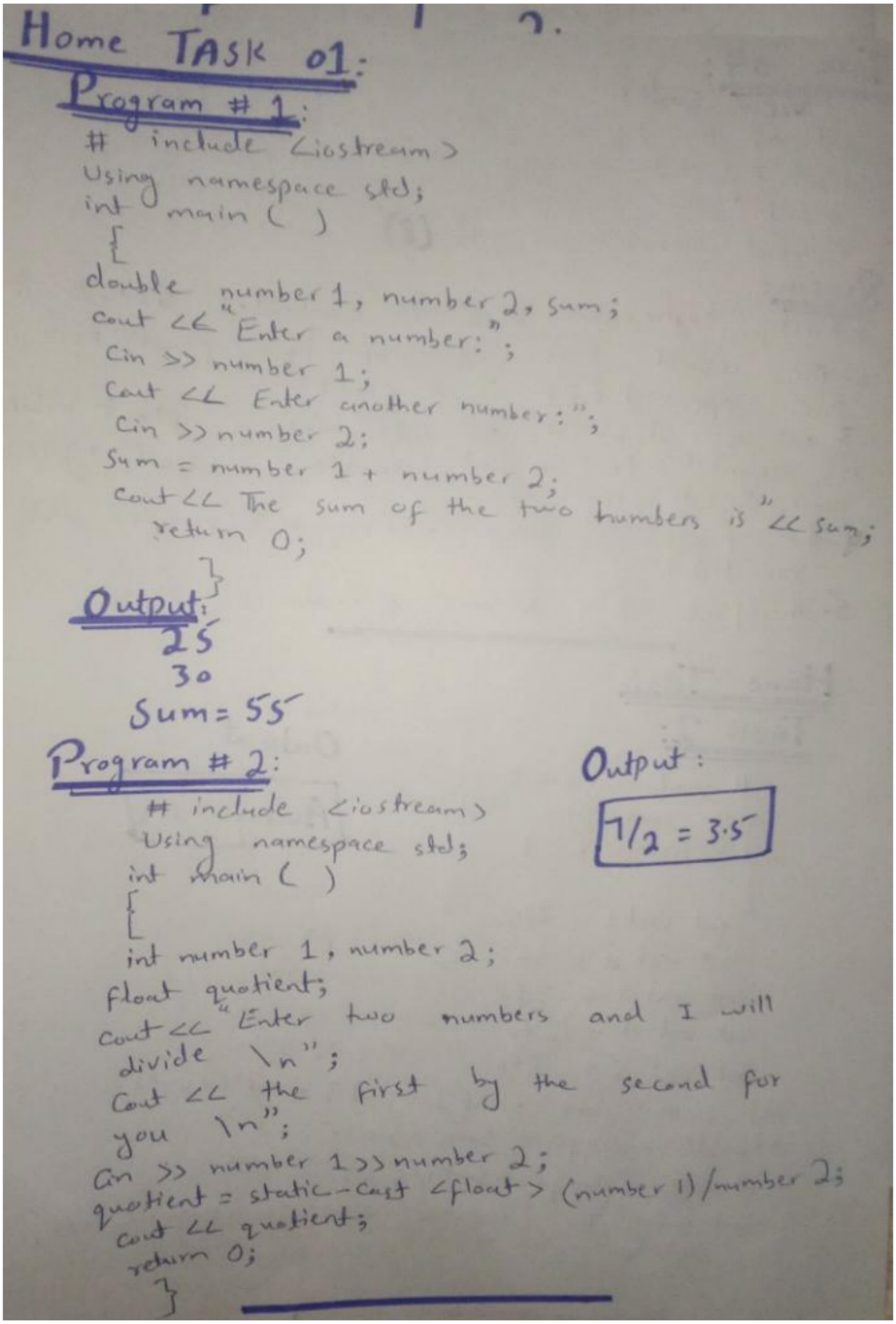
   cin >> number1, number2;

   quotient = float<static_cast>(number1) / number2;

   cout << quotient
}
```

Handwritten code:

HOME TASK 1 — PROGRAMS



Home TASK 01:

Program # 1:

```
# include <iostream>
Using namespace std;
int main ( )
{
double number 1, number 2, sum;
cout << "Enter a number: ";
Cin >> number 1;
Cout << Enter another number: ";
Cin >> number 2;
Sum = number 1 + number 2;
Cout << The sum of the two numbers is " << sum;
return 0;
}
```

Output:

25

30

Sum = 55

Program # 2:

```
# include <iostream>
Using namespace std;
int main ( )
{
int number 1, number 2;
float quotient;
cout << "Enter two numbers and I will
divide \n";
Cout << the first by the second for
you \n";
Cin >> number 1 >> number 2;
quotient = static-Cast <float> (number 1)/number 2;
cout << quotient;
return 0;
}
```

Output:

7/2 = 3.5

HOME TASK 1 — CODE & OUTPUT

Code on compiler:

Coding C++
Auto saved at 21:34:18
RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;
    cout << "Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
    cout << "The sum of the two numbers is " << sum;
    return 0;
}
```

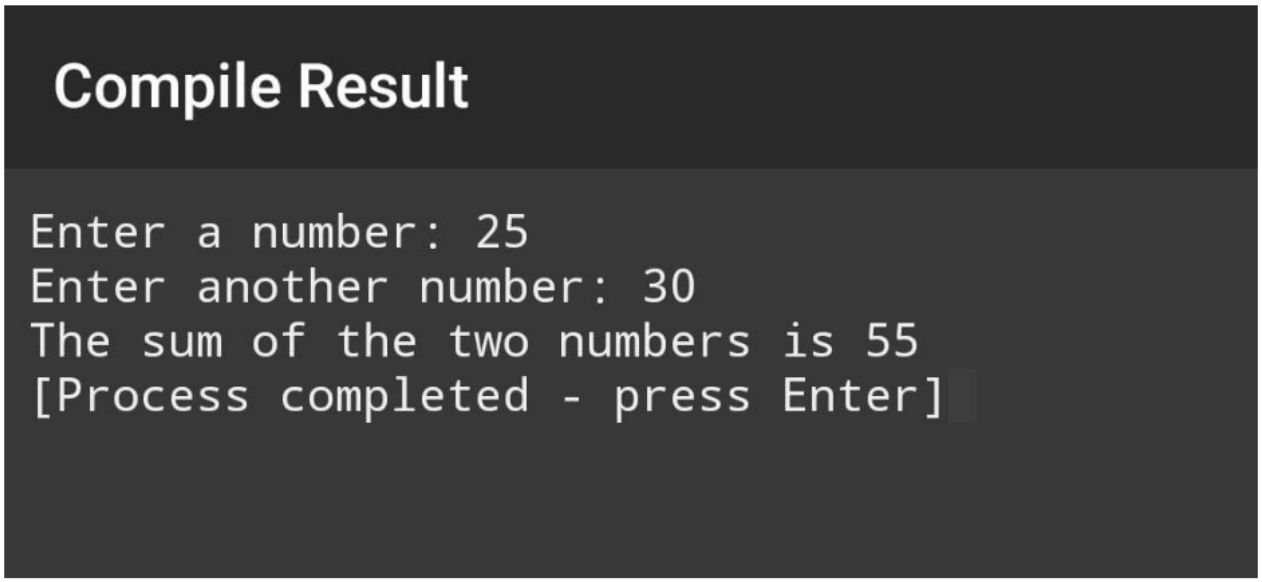
Coding C++
Auto saved at 21:36:09
RUN MENU

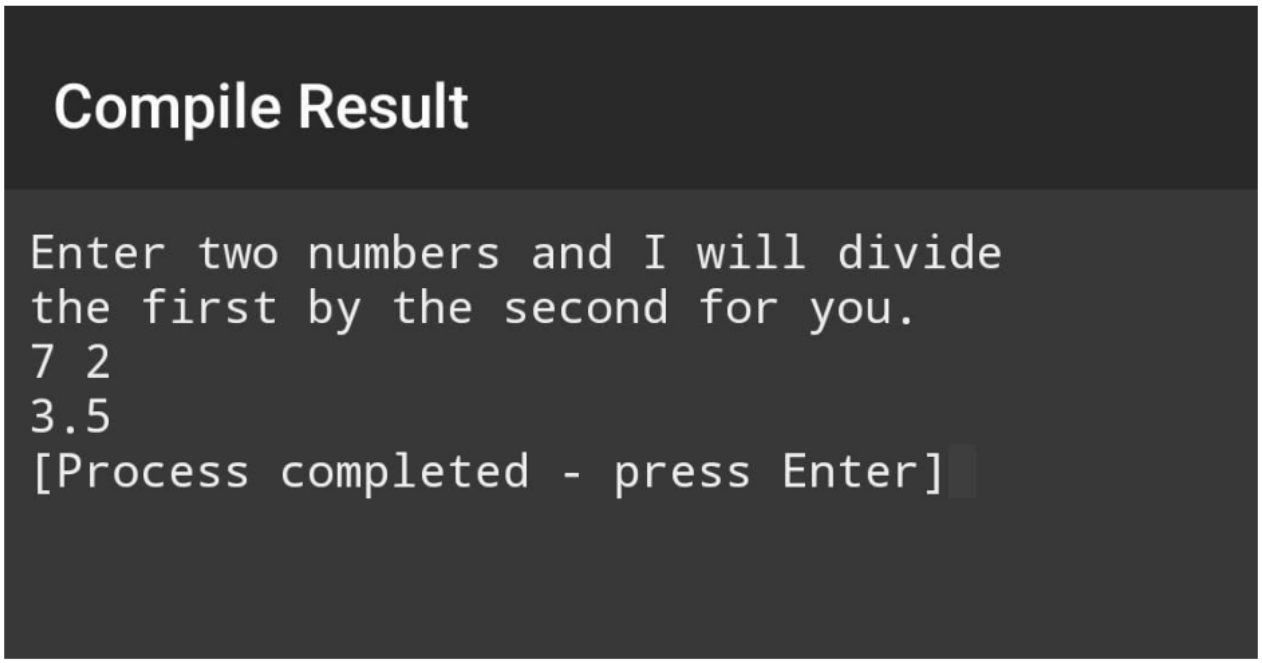
```cpp
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";
    cin >> number1 >> number2;
    quotient = static_cast<float>(number1) / number2;
    cout << quotient;
    return 0;
}
```

HOME TASK 2 — AVERAGE OF VALUES

Output





Home Task 2

Question:
Task 2:

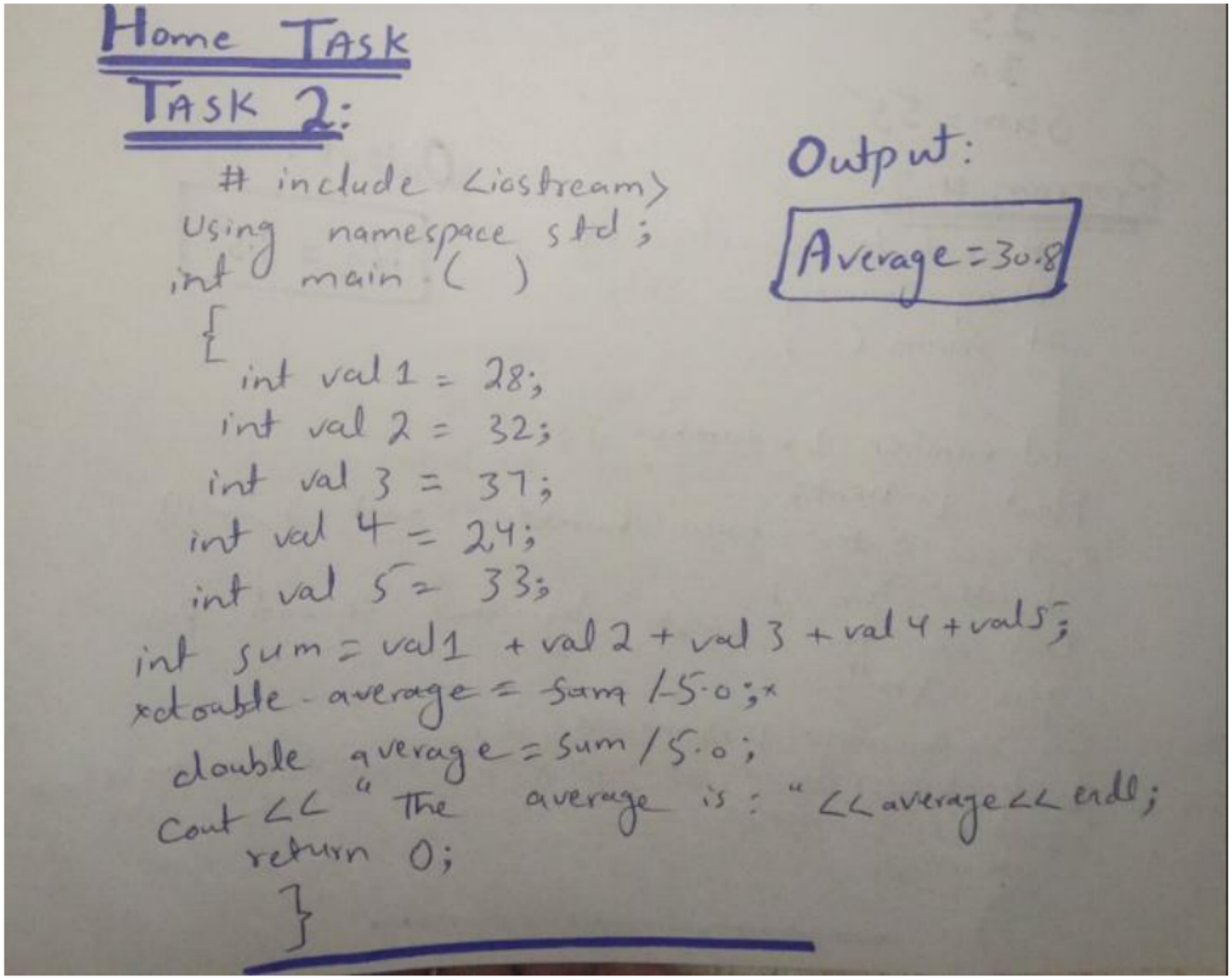
Average of Values

To get the average of a series of values, you add the values up and then divide the sum by the number of values. Write a program that stores the following values in five different variables: 28, 32, 37, 24, and 33. The program should first calculate the sum of these five variables and store

HOME TASK 2 — SOLUTION

the result in a separate variable named sum. Then, the program should divide the sum variable by 5 to get the average. Display the average on the screen.

Handwritten code



Code on compiler

HOME TASK 2 — OUTPUT

Coding C++

Auto saved at 21:40:41 RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1 = 28;
    int val2 = 32;
    int val3 = 37;
    int val4 = 24;
    int val5 = 33;

    int sum = val1 + val2 + val3 + val4 + val5;

    double average = sum / 5.0;

    cout << "The average is: " << average << endl;

    return 0;
}
```

Output

Compile Result

```
The average is: 30.8

[Process completed - press Enter]
```

END OF ASSIGNMENT

End of Assignment